

FSK: Homework Problem Numbers

2025

2 Chapter 2

Chapter 2 Motion in 1 dimension

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

P2.3 (a) 2.30 m/s (b) 16.1 m/s (c) 11.5 m/s

P2.5 (a) -2.4 m/s (b) ≈ -3.8 m/s (c) 4.0 s

P2.11 (a) At 10 s $v = 20$ m/s, at 20 s $v = 5$ m/s (b) 263 m

P2.14 (a) 4.98×10^{-9} s (b) 1.20×10^{15} m/s²

P2.16 (a) 20.0 s (b) 1000 m is needed to land (c) The plane can not land.

P2.23 (a) 3.75 s (b) 5.50 cm (c) 0.604 s or 6.90 s (d) 13.3 cm or 47.9 cm for the two times in (c) (e) Explain

P2.27 7.96 s

P2.29 (a) 10.0 m/s (b) -4.68 m/s

P2.32 (a) See tutor’s memo (b) See tutor’s memo (c) -4 m/s² (d) 32 m (e) 28 m

P2.36 (a) 96 ft/s or 29 m/s (b) 3.07×10^3 ft/s² upward = $93.6g = 936$ m/s² (c) 3.13×10^{-2} s

P2.40 (a) $v_B = \left(\frac{1}{\tan \theta}\right) v$ (b) Explanation